

Answer: D

15 The increase in PE = $\left(\frac{-GM_{Earth}M}{R_{above}}\right) - \left(\frac{-GM_{Earth}M}{R_{surface}}\right)$

$$= GM_{Earth}M \left(\frac{1}{R_{surface}} - \frac{1}{R_{above}} \right)$$

$$= (6.67 \times 10^{-11})(5.98 \times 10^{24})(810) \left(\frac{1}{(6.370 \times 10^6)} - \frac{1}{(6.370 + 0.0920) \times 10^6} \right)$$

$$= (3.231 \times 10^{17})(2.235 \times 10^{-9}) = 7.22 \times 10^8 \text{ J}$$

Answer: C

16 The resonating air column in a clarinet produces pleasant music while the other options